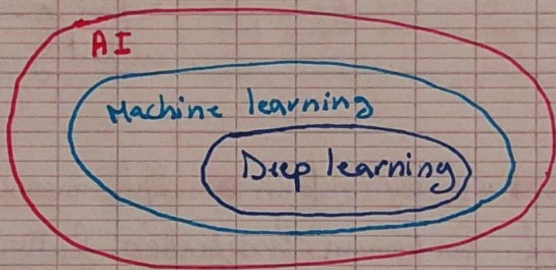
CS229 : Intro to machine learning

(Lecture 1)

Essential takeaways

1. ML systems improve through experience rather than require every behavior to be manually programmed.
2. A clear ML problem should define its experience, task, & performance measure.
3. Supervised learning uses labels, unsupervised learning does not.
4. Regression predicts continuous values, while classification predicts categories.
5. Reinforcement learning teaches behavior through reward signals.
6. Deep learning is only one subset of machine learning.



What is machine learning?

- **Arthur Samuel's definition:** (1959) Machin learning gives computers th ability to learn without being explicitely programmed for every situation.

- **Tom Mitchell's definition :** A program learns form experience E, with respect to a task T, measured using a performance P.

Supervised learning

* In supervised learning, the training data contains:
  - input x
  - correct output or label y

The algo learns a function $f(x) \to y$

It can then predict y for a new input it has not seen before.

* **Regression:**
  - Predicts a continuous numerical value.
  - Example: input x → size of a house
    output y → estimated price
  - A simple model might fit a straight line through the training example. A more complex one could fit a curve.

* **Classification:**
  - Predicts one of a discrete set of categories ≥ 2
  - Example: input x → info about a tumor
    output y → Benign / Malignant

* An input usually contains multiple measurements called features

Example: For tumour classification, features might be:

- Tumor size
- Patient age
- Cell size
- ⋮

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Input can be represented as a feature vector

* **Some algorithms:**
  - Linear regression: predicts continuous values
  - Logistic regression: performs classification
  - Support vector machines: find boundaries between classes
  - Kernels: allow an algo to work implicitely in infinite feature spaces
  - Neural networks: learn complex mapping between inputs & outputs.
  - Gradient descent & backpropagation: methods used to train models.

Unsupervised learning

* In unsupervised learning, the data contains inputs with no labels
$\{ x^{(1)}, x^{(2)}, \dots, x^{(n)} \}$

* Clustering:
  - Groups similar examples together
  - K-means is a common clustering algo.
  - Cocktail party problem:
    - ↳ Goal is to separate recording into individual speakers without being given clean examples of each voice.
    - ↳ Can be addressed using independent component analysis (ICA)
  - Learning from text:
    - ↳ without labels, algo can discover relationships such as "Man" relates to "woman" like "king" & "queen".
    - ↳ The structure is learned from patterns in the raw text itself.

Reinforcement learning

* In reinforcement learning, an agent interacts with an environment:
  1. It observes the current situation
  2. It chooses an action
  3. It receives a reward / penalty
  4. It learns which behavior produces greater reward